ICC-1 Hematology: high yield

Year 3, 2026-2027. One page per lecture, then one summary per Specification Grid topic.

Part 1: lectures

1. Thrombophilia, inherited and acquired [008b, 008c]
2. Haemostasis [008a] (revision notes and LGW)
3. DVT and pulmonary embolism [017]
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Part 2: topic summaries

1. Thrombophilia, DVT and pulmonary embolism
2. Anaemias (megaloblastic, iron deficiency, haemolytic)

PART 1: LECTURE

Thrombophilia, inherited and acquired [008b, 008c]

Topic: Thrombophilia, DVT and pulmonary embolism

Basics

- Thrombophilia = inherited or acquired tendency to venous thrombosis. Virchow's triad: stasis, vessel wall damage, blood changes. In VTE, coagulability and stasis matter most.
- Provoked VTE (transient, removable risk factor) = lower recurrence; unprovoked (no factor, or persistent factor) = high recurrence.

Inherited

- Antithrombin deficiency: AD, recurrent VTE < 40, pregnancy VTE; highest risk of the natural anticoagulant defects. Clue = heparin resistance (no APTT prolongation on heparin). Acquired low AT: nephrotic syndrome, DIC, severe liver disease.
- Protein C deficiency (AD): VTE in heterozygous adults; homozygous = neonatal purpura fulminans/DIC. Warfarin skin necrosis: protein C (half-life ~6 h) falls before II, VII, IX, X.
- Protein S: cofactor for APC; free PS (40%) active, C4b-bound (60%) inactive.
- Factor V Leiden = most common: G1691A, Arg506Gln, partial resistance to APC; mainly venous. ~92% of APC resistance.
- Prothrombin G20210A: raised prothrombin level, at least 2-fold VTE risk.
- Hyperhomocysteinaemia: arterial and venous thrombosis; acquired causes B12/folate deficiency, smoking, renal damage, drugs, age.

Acquired

- Antiphospholipid syndrome: thrombosis and/or pregnancy loss. Lupus anticoagulant prolongs APTT in vitro (binds reagent phospholipid), not corrected by normal plasma, yet causes thrombosis.
- Sapporo: ≥ 1 clinical + ≥ 1 lab criterion. Clinical: thrombosis; fetal death ≥ 10 wk; premature birth ≤ 34 wk; ≥ 3 miscarriages < 10 wk. Lab: LA, anticardiolipin, anti-beta2 GPI; positive twice ≥ 12 weeks apart.
- HITT: IgG to PF4-heparin, platelet FcγRIIa activation; platelets fall 4-10 days after unfractionated heparin. Check baseline platelets and APTT before heparin.
- Myeloproliferative: polycythaemia (viscosity); essential thrombocythaemia (platelets > 1000).

When and how to test

- Suspect if: VTE < 40-45 y, arterial thrombosis < 30 y, unusual site (cerebral vein), recurrent VTE, strong family history, ≥ 3 miscarriages.
- Basic tests (CBC, PT, APTT, TT, fibrinogen, D-dimer) and history rule out most acquired causes. D-dimer: high sensitivity, low specificity.
- Test inherited thrombophilia if young (< 40), thrombosis-prone family, or result changes treatment (pregnancy, children).
- PCR panel: FV G1691A, prothrombin G20210A, MTHFR C677T/A1298C.
- Timing: at least 1 month after anticoagulation ends; avoid pregnancy, pill/HRT, illness. DNA tests any time.

Source: ICC-1 [008]b, [008]c slides; [008] thrombophilia student notes

PART 1: LECTURE

Haemostasis [008a] (revision notes and LGW)

Topic: Thrombophilia, DVT and pulmonary embolism

Platelets

- Count 150-410 x10^9/L, lifespan 10 days, anucleate discs; made by megakaryocyte fragmentation; thrombopoietin is the main regulator.
- High shear: GPIb-IX-V binds VWF, then firm adhesion. Low shear: GPIa/IIa binds collagen directly.
- Aggregation: GPIIb/IIIa + fibrinogen bridges (defective in thrombasthenia). Also mediates clot retraction.
- Dense granules: ADP, ATP, 5-HT, calcium. Alpha granules: PF4, PDGF, fibrinogen, VWF. TXA2 = main amplification loop and vasoconstrictor.
- Inhibitors of platelets: endothelial prostacyclin (PGI2) and nitric oxide.

Coagulation

- Extrinsic tenase: TF + VIIa + PL + Ca -> Xa (initiation). Intrinsic tenase: IXa + VIIIa + PL + Ca -> Xa. Prothrombinase: Xa + Va + PL + Ca -> thrombin burst.
- TFPI switches off initiation (complex with VIIa, TF, Xa).
- All enzymes are serine proteases except XIII (transglutaminase); XIIIa cross-links fibrin.
- Contact factors (XII, prekallikrein, HMWK): deficiency prolongs APTT but no bleeding. XI deficiency may bleed at surgery.
- Vitamin K-dependent: II, VII, IX, X, protein C, protein S (gamma-carboxylation). Warfarin = vitamin K antagonist. Deficiency causes: obstructive jaundice, antibiotics, malabsorption.

Natural anticoagulants and fibrinolysis

- Antithrombin: main inhibitor of thrombin and Xa; action enhanced by heparin.
- Thrombin-thrombomodulin activates protein C; APC + protein S inactivate Va and VIIIa.
- t-PA converts fibrin-bound plasminogen to plasmin. PAI inhibits t-PA; alpha2-antiplasmin inhibits circulating plasmin. Therapeutic t-PA: MI, stroke, massive PE.

Sequence and tests

- After injury: vasoconstriction -> platelet contact -> platelet plug -> fibrin clot -> fibrinolysis.
- Bleeding time 3-8 min; prolonged in thrombocytopenia, platelet defects (aspirin), VWD, vascular defects.
- PT (VII, X, V, II, I): monitors warfarin. INR = (patient PT / mean normal PT)^ISI.
- APTT 26-40 s (VIII, IX, XI, XII + common): monitors unfractionated heparin.
- Isolated long PT = factor VII; isolated long APTT = VIII, IX, XI or XII.
- TT: affected by fibrinogen, heparin, FDPs.

Source: ICC-1 [008]a haemostasis student notes; ICC1[008] Haemostasis LGW

PART 1: LECTURE

DVT and pulmonary embolism [017]

Topic: Thrombophilia, DVT and pulmonary embolism

DVT: diagnosis

- VTE = DVT and/or PE. Risks: surgery, immobility, age > 60, obesity, pregnancy, pill/HRT, cancer, prior VTE, thrombophilia, dehydration.
- Signs (pain, unilateral swelling, warmth, calf tenderness) are not reliable. Differentials: cellulitis, ruptured Baker's cyst, calf haematoma, muscle tear, lymphoedema.
- Three steps: clinical probability (Wells), imaging, D-dimer.
- Compression duplex ultrasound = investigation of choice (less sensitive for calf). Venography = old gold standard; CT venography for iliac/IVC.
- D-dimer: high sensitivity, low specificity; value = negative result with low pre-test probability.

DVT: treatment

- Uncomplicated: home treatment. Admit for limb-threatening DVT, high bleeding risk, IV analgesia, CrCl < 30, HIT history.
- DOACs preferred over VKAs; if warfarin, overlap with LMWH/UFH.
- No routine thrombolysis; consider (catheter-directed) for phlegmasia cerulea dolens or young iliofemoral DVT.
- IVC filter only if anticoagulation contraindicated; retrievable, remove later. Routine stockings not recommended.
- Duration: provoked 3-6 months then stop; unprovoked or persistent risk (cancer) 3-6 months then indefinite if bleeding risk low-moderate.
- Complications: PE (30-50% of proximal DVT), post-thrombotic syndrome (valve damage), phlegmasia, paradoxical embolism (ASD).
- Upper limb DVT: catheters, thoracic outlet, Paget-Schroetter (effort); ultrasound.

PE: diagnosis

- RV strain -> low LV filling -> hypotension/shock. Symptoms: dyspnoea, pleuritic pain, haemoptysis, syncope. Commonest sign: tachypnoea.
- Modified Wells: > 4 likely (go to imaging); ≤ 4 unlikely (D-dimer first).
- CTPA = main test; better with cardiorespiratory disease. V/Q if CTPA unsuitable; normal V/Q virtually excludes PE; hard to read if CXR abnormal/COPD.
- ECG: commonest sinus tachycardia; RBBB, RAD, S1Q3T3 less common; normal ECG does not exclude. ABG: hypoxaemia, hypocapnia.
- Troponin, BNP and echo = risk stratification, not diagnosis.

PE: treatment

- Severity categories A-E (2026 AHA/ACC): A-B early discharge; C-D admit; E critical care and reperfusion.
- Reperfusion (systemic or catheter thrombolysis, thrombectomy, embolectomy) for E and selected D only.
- LMWH over UFH when parenteral needed; DOACs over VKAs. Duration 3-6 months, extended if unprovoked or persistent risk.

Source: ICC-1 [017] DVT and pulmonary embolism slides

PART 1: LECTURE

Approach to anaemia [052]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Definition and mechanisms

- Anaemia = Hb below normal for age and sex: men < 13, women < 12, pregnancy < 11, 6-12 y < 11.5, birth < 14 g/dL.
- Mechanisms: reduced marrow production, increased destruction (haemolysis), loss (bleeding).
- Dehydration can mask anaemia; pregnancy and splenomegaly (bigger plasma volume) can mimic it.

Classification

- By red cell size (usual first step):
- Microcytic (MCV < 80, MCH < 27): iron deficiency, thalassaemia, some chronic disease, lead, sideroblastic.
- Normocytic: many haemolytic anaemias, chronic disease, acute blood loss, renal disease, mixed deficiency, marrow failure.
- Macrocytic (MCV > 100): megaloblastic (B12, folate) or non-megaloblastic (alcohol, liver disease, hypothyroidism, myelodysplasia, aplastic anaemia, reticulocytosis).
- Leucoerythroblastic film (normoblasts + myelocytes): marrow infiltration; also severe bleeding, acute haemolysis.
- Mixed iron + folate/B12 deficiency: indices may be normal, film dimorphic.

Clinical

- Adaptation: higher cardiac output, flow to brain and heart, raised 2,3-DPG (right shift).
- Severity of symptoms depends on Hb level, speed of onset and comorbidity.
- Acute bleeding: Hb may be near normal early; manage by blood loss and cardiovascular state.
- Mucous membrane pallor is more reliable than skin pallor; hyperdynamic circulation; heart failure if severe.
- History clues: menorrhagia, bowel change in elderly (colorectal cancer), bone pain (myeloma), arthralgia (chronic disease), diet, drugs, gastrectomy.

Investigations

- FBC indices, other cell lines (pancytopenia = marrow problem or hypersplenism), blood film, reticulocytes, haematinics, U&Es, LFTs.
- Reticulocytes 0.5-2.5% (25-125 x10^9/L). High in chronic haemolysis and after bleeding (rise 2-3 days, peak 6-10 days). Low = marrow failure or lack of erythropoietin (CKD).
- Bone marrow not always needed. Aspirate: cell detail, M:E ratio, iron stain. Trephine: cellularity, fibrosis, infiltrates.
- Further tests by cause: haemolysis screen (DAT), endoscopy/colonoscopy, imaging, electrophoresis for myeloma.

Treatment

- Find the cause first; treat the patient, not the Hb. Mild anaemia often needs no correction; sickle cell or MDS patients tolerate low baseline Hb (treat only a significant fall).
- Transfusion only if Hb very low, symptomatic or acute bleeding; it is not a cure.

Source: ICC-1 [052] lecture and student notes; haematology anaemia handout

PART 1: LECTURE

Iron deficiency anaemia [069a]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Iron basics and stages

- Body iron 40-50 mg/kg; ~75% in haemoglobin. No physiological way to excrete iron: extra loss = bleeding.
- Absorption: haem > Fe2+ > Fe3+. Reduced by high gastric pH (gastrectomy, vagotomy, H2 blockers, calcium antacids), coeliac, Crohn's, phytates, tannins; helped by vitamin C (ascorbate).
- Order of deficiency: stores depleted -> film/marrow changes -> anaemia (last). Treat even if Hb normal.
- On treatment the order reverses: Hb normalises before stores refill, so continue iron for several months.

Causes

- Chronic blood loss: menorrhagia (commonest in UK), GI loss (peptic ulcer, gastric/colon cancer), hookworm (commonest worldwide), intravascular haemolysis (haemosiderinuria), blood donation.
- Poor intake or absorption (coeliac, gastrectomy); increased demand (pregnancy, prematurity, growth spurt).
- Older patient or family history of bowel cancer with unexplained IDA: exclude malignancy.

Clinical features

- Tiredness, exertional dyspnoea; angular stomatitis, koilonychia (spoon nails), glossitis, pharyngeal web dysphagia (rare), irritability, poor cognition in children.

Investigations

- FBC: low MCV, MCH, MCHC; thrombocytosis common. Film: microcytic hypochromic cells, pencil cells, target cells.
- Iron studies: LOW iron, HIGH TIBC, LOW ferritin; transferrin saturation (iron/TIBC x 100) < 15%.
- Ferritin is an acute phase protein: may be normal/high in acute illness despite deficiency.
- Soluble transferrin receptor rises early in IDA and separates IDA from ACD (not routine).
- Perls' Prussian blue marrow stain: reliable but invasive, rarely needed.
- Find the cause: coeliac screen (anti-tTG, duodenal biopsy), gastroscopy/colonoscopy, capsule endoscopy, stool for ova/parasites, gynae assessment.

Treatment and differentials

- Ferrous sulphate 200 mg three times daily for 4-6 months, plus treat the cause. Side effects: bloating, constipation, diarrhoea.
- Poor response: poor compliance (commonest), malabsorption, wrong diagnosis (thalassaemia trait), ongoing bleeding.
- IV iron if oral iron not tolerated or ineffective: whole dose in a day, but Hb rises at the same rate.
- Thalassaemia trait: microcytosis with normal iron studies; Hb usually > 9; HbA2 raised.
- ACD (e.g. RA): LOW iron, LOW TIBC, normal/HIGH ferritin; iron trapped in macrophages.
- Sideroblastic: ring sideroblasts in marrow; MCV low, normal or high; primary (MDS) or alcohol, drugs, toxins.

Source: ICC1 [069] Iron deficiency anaemia slides; ICC-1 [052, 069] lecture notes

PART 1: LECTURE

Megaloblastic anaemia [069b]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Basics

- Megaloblastic = macrocytic anaemia (MCV > 100 fl) from failed DNA synthesis; main causes B12 and folate deficiency.
- Non-megaloblastic macrocytosis: alcohol, liver disease, hypothyroidism, aplastic anaemia, reticulocytosis, cytotoxic drugs (hydroxyurea), myelodysplasia.

Vitamin B12

- Animal foods only (eggs, dairy, fish); stores 3-4 mg in liver last about 3 years, so deficiency is slow.
- Freed in stomach, binds intrinsic factor (gastric parietal cells), absorbed in terminal ileum, carried by transcobalamin II.
- Needed to make methionine from homocysteine (activates folate for DNA) and methylmalonyl CoA -> succinyl CoA.
- Causes: vegan diet; pernicious anaemia (antibodies to IF and/or parietal cells); ileal disease/ resection; gastrectomy; bacterial overgrowth; alcohol; tropical sprue; fish tapeworm; repeated nitrous oxide; transcobalamin II deficiency.
- Neurology even without anaemia: peripheral neuropathy, subacute combined degeneration, optic atrophy, dementia.

Folate

- Green vegetables, liver, beans; stores about 10 mg last only about 4 months; heat labile.
- Absorbed in duodenum and jejunum; needs B12 to become active THF (MTHFR step).
- Causes: poor diet (elderly, alcohol, poverty), coeliac, pregnancy, chronic haemolysis, malignancy, psoriasis; drugs: anticonvulsants, sulphasalazine, methotrexate, trimethoprim, pyrimethamine; dialysis.
- Deficiency causes neural tube defects; adult neurology less common.

Investigations

- FBC: macrocytosis, anaemia may be absent, pancytopenia if severe.
- Film: oval macrocytes and hypersegmented neutrophils (> 5% with > 5 lobes, "shift to right") appear early, even before anaemia; tear drop cells; low reticulocytes.
- Raised unconjugated bilirubin and LDH = ineffective erythropoiesis.
- Serum B12 + folate together; red cell folate best for stores. IF antibodies specific for PA; parietal cell antibodies are not.
- Bone marrow (hypercellular, megaloblasts) usually not needed.

Treatment

- Hydroxocobalamin 1 mg IM 3x weekly for 2 weeks, then 1 mg IM 3-monthly for life if malabsorption; oral cyanocobalamin if dietary.
- Never give folic acid before B12 (can worsen neurology). Treat promptly.
- Severe anaemia: short course of folate and iron may be needed as new red cells surge.
- Folate deficiency: folic acid 5 mg daily for 3-4 months; 400 µg daily before conception prevents NTDs.

Source: ICC-1 [069] Deficiency anaemias part 2, megaloblastic anaemia

PART 1: LECTURE

Autoimmune haemolytic anaemia [077]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Types

- Haemolytic anaemia = shortened red cell lifespan. Inherited: membrane (spherocytosis), enzyme (G6PD), Hb (sickle, thalassaemia). Acquired: immune or non-immune.
- Warm AIHA: IgG +/- complement, active at 37 C; spherocytes; mostly extravascular. Causes: idiopathic, SLE, RA, lymphoma/CLL, HIV, drugs.
- Drugs: methyldopa = true AIHA; penicillin = antibody to drug on red cell; rifampicin = complement via immune complex.
- Cold AIHA: active at room temperature or below, in cool extremities; complement-mediated, mostly intravascular; film agglutination. Causes: Mycoplasma, EBV, lymphoproliferative disease.
- Alloimmune: delayed transfusion reaction (extravascular), ABO-incompatible transfusion (intravascular), haemolytic disease of newborn.
- Non-immune: MAHA with schistocytes (HUS, TTP, DIC, HELLP, vasculitis), mechanical valves/ grafts, malaria, clostridia, burns, PNH, march haemoglobinuria.

Clinical

- Anaemia, jaundice, splenomegaly, pigment gallstones (chronic), plus features of the cause.

Four questions in investigation

- 1. Is it haemolysis? Raised reticulocytes (polychromasia), raised unconjugated bilirubin, raised LDH, low haptoglobin, raised urinary urobilinogen; marrow erythroid hyperplasia (rarely needed).
- 1. Immune or not? DAT (direct Coombs): positive = immune, negative = non-immune. Indirect test = free antibodies, used before transfusion.
- 1. Where? Extravascular = macrophages, mainly spleen. Intravascular = haemoglobinaemia, haemoglobinuria, haemosiderinuria, very low haptoglobin, very high LDH, methaemalbumin (Schumm's test).
- 1. Cause: haemolysis days after transfusion + DAT positive = delayed reaction (new alloantibody); AIHA + nodes + lymphocytosis = lymphoid malignancy; fever + agglutination = cold AIHA (Mycoplasma IgM).
- TTP: fever, neurology, renal impairment, low platelets, fragments; low ADAMTS13 is diagnostic.
- PNH: anaemia, abdominal pain, thrombosis, intermittent haemoglobinuria.

Management

- Folic acid 5 mg daily (high turnover uses folate).
- Transfuse only if severe, life-threatening or troublesome.
- Treat the specific cause.

Source: ICC-1 [077] Haemolytic anaemia

PART 1: LECTURE

Haemoglobin assay and blood indices [060] (prelab and lab)

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Sample and safety

- Gloves (peel off wrist to fingers), buttoned lab coat; sharps to sharps bin, disposables to red biohazard bag; spills: 10% hypochlorite for 10 min.
- CBC sample: EDTA, well mixed, run within 6 h (or store up to 24 h at 4 C).

Cyanmethaemoglobin method (ICSH reference)

- Drabkin's solution: red cells lyse; potassium ferricyanide oxidises Hb to methaemoglobin; potassium cyanide converts it to cyanmethaemoglobin.
- Stand 5-10 min; zero at 540 nm with a blank; absorbance at 540 nm is proportional to Hb.
- Hb = (absorbance test / absorbance standard) x standard concentration; or read from a standard curve made with HiCN standard when many samples.
- Micropipette: fit tip, press to first stop, release to draw up, check bubbles, expel at second stop.
- Check: Hb (g/dl) x 3 is roughly the PCV.

Red cell indices

- MCV = PCV x 10 / RBC (fl); MCH = Hb x 10 / RBC (pg); MCHC = Hb x 100 / PCV (g/dl).
- Uses: classify anaemia (micro-, normo-, macrocytic) and quality control.
- RDW = variation in red cell size (anisocytosis); raised in iron deficiency, megaloblastic anaemia and haemolysis.
- Poikilocytosis = abnormal shapes (on the film).
- Each lab sets its own reference ranges; approximate values: Hb men 13-17, women 12-15 g/dl; MCV 80-100 fl; MCH 27-32 pg.

Physiological variation

- Highest Hb at birth. Pregnancy: plasma expands more than red cells, lower limit Hb 11 g/dl.
- High altitude and smoking (carboxyhaemoglobin) raise Hb and PCV.

Other CBC points

- Reticulocytes 0.5-2.5%: reflect marrow activity; mature in 2-3 days (1 day in blood).
- Shift to left (myelocytes, bands > 6%): severe infection, marrow infiltration.
- Nucleated red cells beyond the neonatal period = marrow stress or damage; leukoerythroblastic = immature red and white cells.
- Low Hb, WBC and platelets together = pancytopenia.

Source: Hb estimation assay; ICC-1 [044] prelab; [060] Hb assay checklists

PART 1: LECTURE

Blood film preparation [066] (prelab and lab)

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Lab safety

- Gloves before handling blood; remove them before touching notebooks or microscopes and before leaving the lab; peel from wrist to fingers.
- Sharps and slides go in the sharps bin; disposables in red biohazard bags.
- Blood spill: pour 10% sodium hypochlorite, cover with absorbent towel, leave at least 10 min, repeat, then wash.
- Skin penetrated: wash with soap and water and encourage bleeding. Eyes: rinse gently with water or saline. Mouth: spit and rinse. Report at once.

Making the film

- Sample: EDTA blood or finger/heel prick (choice depends on the case).
- Drop (about 20 uL) near the end of a clean slide; pusher at 30-45 degrees (checklist: 30-40); pull back until blood spreads across the width, then push forward quickly and smoothly.
- Good film covers 2/3 to 3/4 of the slide, smooth, no irregularities. Wrong angle pushes large cells (monocytes) to the tail and spoils the differential.
- Air dry, stain (Wright-Giemsa most common, or Leishman), wash, air dry.

Examining the film

- x10 first: large abnormal cells (blasts) and RBC distribution. Then x40 and x100 (oil): RBC morphology, WBC differential, platelets.
- Good stain: RBCs pink-salmon; neutrophil granules lilac; eosinophil red-orange; basophil dark blue-black; clean background.
- Normal RBC: biconcave, anucleate, central pallor, 7-8 um, MCV 80-100 fL, MCHC 31-37 g/dL. Platelets 150-450 x10^9/L, 1-4 um.

Abnormal red cells and their links

- Target cells: thalassaemia. Spherocytes (no central pallor): hereditary spherocytosis. Tear drop cells: myelofibrosis, thalassaemia major.
- Acanthocytes: abetalipoproteinaemia. Basophilic stippling: lead. Schistocytes (fragments): microangiopathic haemolysis. Agglutination: cold agglutinins.
- Elliptocytes (pencil cells): hereditary elliptocytosis; also increased in iron deficiency.
- B12 deficiency: macrocytes (MCV > 100) and hypersegmented neutrophils.
- Leuko-erythroblastic film: normoblasts + myelocytes.

Reticulocytes and marrow iron

- Young RBCs with RNA remnants; supravital stain (new methylene blue or brilliant cresyl blue). Normal 0.5-2.5%.
- High in chronic haemolysis and after bleeding (rise 2-3 days, peak 6-10 days). No rise in anaemia = poor marrow or no erythropoietin.
- Prussian blue marrow: iron deficiency = no iron anywhere; chronic disease = iron in macrophages, none in red cell precursors.

Source: [066] prelab session; [066] film benchaid; [066] revision benchaid; blood film checklist

PART 1: LECTURE

Haematology cases LGW [092]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Thrombophilia work-up (young DVT, family history)

- Delay thrombophilia tests until at least 1 month after anticoagulation ends: the acute clot and anticoagulants distort results, and results do not change acute management.
- Screen: PT, APTT, TT, protein C, protein S, antithrombin. If all normal: PCR panel for factor V Leiden (G1691A), prothrombin G20210A, MTHFR.
- Factor V Leiden: autosomal dominant; Arg506Gln makes factor Va resistant to activated protein C (APC resistance).
- Risk of venous thrombosis about 4-fold (heterozygous) and about 20-fold (homozygous). Offer screening to affected relatives.

Antiphospholipid syndrome (recurrent miscarriage)

- Three consecutive early miscarriages with no anatomical or hormonal cause: weakly positive lupus anticoagulant suggests APS but is not diagnostic.
- Next: anticardiolipin and anti-beta-2 glycoprotein 1 antibodies.
- Repeat after 12 weeks; still positive with high titre = diagnostic.
- Confirmed APS: anticoagulant therapy as soon as she becomes pregnant.

Autoimmune haemolytic anaemia

- Normocytic normochromic anaemia + jaundice = suspect haemolysis.
- Confirm haemolysis: bilirubin up, haptoglobin down. Reticulocytes up (13%) = effective marrow response.
- Haemoglobinuria (pink-red urine) with high bilirubin and low haptoglobin = intravascular haemolysis.
- Film: spherocytes and red cell agglutination (cold agglutinins). Direct antiglobulin test (DAT) positive is diagnostic of AIHA.
- Cold type antibodies are usually IgM; about half are secondary (lymphoma, viral infection).

TTP in pregnancy

- Fever + neurological change + renal impairment + low platelets + fragmented red cells (microangiopathic haemolysis).
- Normal clotting excludes DIC; normal liver tests exclude HELLP; normal BP argues against pre-eclampsia.
- Cause: antibodies to ADAMTS13, so very large VWF multimers circulate; rare inherited ADAMTS13 deficiency.
- Treatment: early daily plasma exchange with FFP. Untreated mortality high; relapse 30-60%, mostly in the first month. Newer: rituximab, recombinant ADAMTS13.

Source: ICCA [092] haematology cases LGW Qs and As

PART 1: LECTURE

Clinical chemistry tutorial [027]

Topic: Anaemias (megaloblastic, iron deficiency, haemolytic)

Lipid profile

- Reference (fasting): total cholesterol < 5.2 mmol/L (< 200 mg/dL); TG < 1.7 mmol/L (< 150 mg/dL); HDL-C > 1 mmol/L (> 40 mg/dL).
- Convert: cholesterol and HDL-C mg/dL = mmol/L x 38.7; TG mg/dL = mmol/L x 88.57.
- Case: cholesterol 6.7 (259 mg/dL) high, TG 1.4 (124) normal, HDL-C 0.83 (32) low = hypercholesterolaemia with normal TG.

Secondary causes (exclude before treating)

- High cholesterol (LDL): hypothyroidism, chronic liver disease, nephrotic syndrome; progestins, thiazides, glucocorticoids, ciclosporin.
- High TG: obesity, type 2 DM, pregnancy, acute hepatitis; alcohol, oestrogens, antifungals, isotretinoin.
- Low HDL: malnutrition, obesity, metabolic syndrome; anabolic steroids, tobacco.

Assessment and management

- Look for familial hypercholesterolaemia signs (tendon xanthomata, arcus, xanthelasma); absence does not exclude FH.
- History: smoking, alcohol, DM, drugs, past PVD or stroke, family history of premature CVD. Exam: BMI, waist, BP, xanthomata, bruits, pulses.
- Manage: exclude secondary causes, treat hypertension, weight loss with dietician, then estimate 10-year risk to decide on drugs.
- QRISK2: UK algorithm for 10-year CVD risk, preferred by NICE since 2010; calibration varies in other populations. Lifestyle and other risk factors first before statins.
- New lab method: convert to reference-method value by regression (HDL-C 0.83 = 1.0 mmol/L; QRISK 8.07% to 6.7%). Method can change the statin decision.

Cardiac markers

- Typical pain + normal ECG does not exclude MI (ECG not always positive).
- Total CK is not heart-specific. Troponin is heart-specific but rises 4-8 h after onset: negative at 3 h does not exclude MI; repeat later.
- CK-MB also rises at 4-8 h; also in skeletal muscle; CK-MB/total CK > 5% supports MI.
- At 12 h: raised troponin T confirms myocardial damage.
- Re-infarction: troponin stays raised > 1 week, so not useful; CK-MB normalises in 2-3 days, so a new rise (ratio > 5%) = new or extended infarct.

Source: ICC-1 [027] clinical chemistry tutorial Q and answers

PART 2: TOPIC SUMMARY

Thrombophilia, DVT and pulmonary embolism

How clotting is balanced

- Platelets: adhesion (GPIb to VWF), aggregation (GPIIb/IIIa with fibrinogen), amplification by TXA2. Coagulation: tissue factor + VIIa starts it; prothrombinase (Xa + Va) gives the thrombin burst; XIIIa cross-links fibrin.
- Natural brakes: antithrombin (thrombin and Xa, boosted by heparin); protein C + protein S (inactivate Va and VIIIa); TFPI; fibrinolysis (t-PA makes plasmin).
- Tests: PT (VII, X, V, II, I) monitors warfarin; APTT monitors unfractionated heparin; isolated long APTT = VIII, IX, XI or XII (XII causes no bleeding).

Thrombophilia at a glance

Condition	Key fact	Clue
Factor V Leiden	Commonest; Arg506Gln, APC resistance	Young venous clot, family history
Prothrombin G20210A	High prothrombin level	DNA test
Antithrombin deficiency	Highest risk natural anticoagulant defect	Heparin resistance
Protein C deficiency	Short half-life	Warfarin skin necrosis; neonatal purpura fulminans
Antiphospholipid syndrome	Clots and pregnancy loss	Long APTT in vitro but thrombosis; repeat antibodies after 12 weeks
HITT	Antibody to PF4-heparin	Platelets fall 4-10 days after heparin

- Test when: VTE under 40, unusual site, recurrent, strong family history, 3 or more miscarriages. Wait at least 1 month after anticoagulation ends (DNA tests any time).

DVT and PE pathway

- Clinical probability first (Wells): unlikely = D-dimer (a negative result excludes); likely = imaging straight away.
- DVT: compression duplex ultrasound. PE: CTPA (V/Q if CTPA unsuitable). ECG usually sinus tachycardia; S1Q3T3 is uncommon.
- Treat: DOAC preferred (or LMWH overlapping with warfarin). Provoked 3-6 months; unprovoked or cancer = consider indefinite.
- Thrombolysis only for massive PE (shock) or limb-threatening DVT. IVC filter only if anticoagulation is impossible.

PART 2: TOPIC SUMMARY

Anaemias (megaloblastic, iron deficiency, haemolytic)

First step: size of the red cells

Microcytic (MCV under 80)	Normocytic	Macrocytic (MCV over 100)
Iron deficiency, thalassaemia trait, anaemia of chronic disease, sideroblastic, lead	Acute bleeding, haemolysis, chronic disease, renal failure, marrow failure	Megaloblastic (B12, folate); alcohol, liver disease, hypothyroidism, myelodysplasia, reticulocytosis

- Reticulocytes high = bleeding or haemolysis (marrow working); low = marrow failure or no erythropoietin (renal disease).
- Treat the cause, not the number. Transfuse only for severe, symptomatic anaemia or acute bleeding.

Iron deficiency versus look-alikes

	Iron	TIBC	Ferritin	Other
Iron deficiency	Low	High	Low	Pencil cells, raised platelets
Chronic disease	Low	Low	Normal or high	Iron trapped in macrophages
Thalassaemia trait	Normal	Normal	Normal	Raised HbA2, mild anaemia

- Causes: menorrhagia (UK), hookworm (worldwide), GI blood loss. Older patient with unexplained iron deficiency = scope for cancer.
- Ferrous sulphate 200 mg three times daily for 4-6 months, continue after Hb normalises to refill stores. Poor response: compliance first.

Megaloblastic anaemia

- B12: animal foods; stores last about 3 years; needs intrinsic factor, absorbed in terminal ileum. Pernicious anaemia: intrinsic factor antibodies. Neurology (subacute combined degeneration) even without anaemia.
- Folate: stores last about 4 months; poor diet, pregnancy, haemolysis, anticonvulsants, methotrexate.
- Film: oval macrocytes, hypersegmented neutrophils; raised LDH and bilirubin.
- Never give folic acid alone before B12. Hydroxocobalamin IM; folic acid 5 mg daily; 400 µg before conception.

Haemolysis in four questions

- Is it haemolysis? Reticulocytes, unconjugated bilirubin and LDH up; haptoglobin down.
- Immune? DAT (direct Coombs) positive = immune.
- Where? Intravascular: haemoglobinuria, very low haptoglobin. Extravascular: spleen.
- Cause? Warm IgG (SLE, CLL, drugs; spherocytes) versus cold IgM (Mycoplasma, EBV; agglutination). TTP: fever, neurology, renal failure, low platelets, fragments, low ADAMTS13; plasma exchange urgently.

Lab skills

- Hb by cyanmethaemoglobin method, read at 540 nm. MCV = PCV x 10 / RBC; MCH = Hb x 10 / RBC; MCHC = Hb x 100 / PCV.
- Blood film: drop near the end, spreader at about 30-45 degrees, smooth push; read at x10 then oil. Target cells = thalassaemia; spherocytes = hereditary spherocytosis or warm AIHA; schistocytes = MAHA; basophilic stippling = lead.
- Clinical chemistry: lipid units (cholesterol mg/dL = mmol/L x 38.7); troponin rises 4-8 h after MI and stays up over a week, so use CK-MB for re-infarction.